

ASHWIN R

Software Engineer | Full Stack Developer | B.Tech IT Student

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ABOUT

Pre-final year B.Tech IT student at K. Ramakrishnan College of Engineering, Trichy, with some knowledge in designing and building distributed systems, scalable backend services, and full-stack applications. I apply software design principles and data structures to real-world problems — from multi-tenant architectures to authentication infrastructure at institutional scale. Actively seeking opportunities to build high-impact, production-grade software.

EDUCATION

B.Tech — Information Technology | K. Ramakrishnan College of Engineering, Trichy | 2024 – 2028
(Penultimate Year 2026)

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Dart, **CS Fundamentals:** Data Structures, Algorithms, Software Design

Systems & Backend: Distributed Auth, REST APIs, SQL/NoSQL, Docker, OAuth 2.0 **Tools:** GitHub Actions (CI/CD), VS Code, Antigravity

EXPERIENCE

Software Development Intern | National Institute of Technology, Trichy (NIT Trichy)

Java • React 19 • Node.js • OAuth 2.0 • PKCE • SQLite • Distributed Systems

Designed and developed AuthForge — a centralized OAuth 2.0 / OpenID Connect Identity Provider serving NIT Trichy's campus ecosystem, applying distributed systems design to handle concurrent authentication requests across multiple applications.

Implemented the full authorization code flow with PKCE, JWT-based access control, and refresh token rotation — applying software design patterns to build a reliable, scalable auth system.

PROJECTS

Task Flow — Multi-Tenant Distributed Task Management System github.com/Ashwin-R05/Task_Flow

Java • Node.js • MongoDB • JWT • Docker • GitHub Actions • REST API

Architected a scalable multi-tenant system with complete data isolation using a shared-database, shared-collection pattern — designed to handle concurrent requests across multiple independent organizations.

Applied software design principles (role-based access control, token-based auth with bcrypt) and containerized the full stack with Docker; automated deployments via GitHub Actions CI/CD pipeline.

Summarizit — Cross-Platform AI Summarization App github.com/Ashwin-R05/summarizit

Dart • Flutter • REST API • Clean Architecture • Android / iOS / Web / Desktop

Designed and built a cross-platform application using Clean Architecture — separating concerns across data, domain, and presentation layers to produce a maintainable, testable codebase.

Integrated an AI summarization API to process and condense long-form content; ships to 6 platforms (Android, iOS, Web, Windows, Linux, macOS) from a single codebase.

ACHIEVEMENTS

2nd Place — Hackathon, Paavai Institute of Technology

Flutter Web • MERN Stack • Scalable Systems • SDG 11

Awarded 2nd prize for UrbanLens (CityGuard) — a scalable citizen-driven smart city platform built with a 3-member team. Led system architecture, data design, and full presentation — project aligned with UN SDG 11.